

AMOGH SOMISETTY

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EE undergraduate targeting Summer 2027 internships in edge AI, GPU systems, and ML hardware: blind-validated H100 fleet forensics at Princeton, NVIDIA Jetson flight hardware, and funded on-device vision research.

EDUCATION

California Polytechnic State University, San Luis Obispo

B.S. Electrical Engineering GPA: 3.12/4.00

Expected June 2028

San Luis Obispo, CA

- **Coursework:** Computer Architecture, Digital Logic Design, Embedded Systems, Signals & Systems

EXPERIENCE

Creator & Lead Researcher, Theta (open-source GPU reliability)

GPU thermal-power forensics research, advised by ex-Intel silicon-validation faculty

Mar 2026 – Present

San Luis Obispo, CA

- Detected failing GPUs that standard monitoring missed: blind-flagged 3 of **64 H100s** on **Princeton** Research Computing's Della cluster from telemetry alone; the operator independently confirmed two: one RMA-replaced while temperature-indistinguishable from healthy peers, and one re-measured at **+47.8%** thermal resistance with **-2.6%** GFLOPS at matched power.
- Shipped the detection agent behind it (`pip install runtheta`): peer-relative thermal resistance ($R_\theta = \Delta T/P$) by two-way median polish over position- and power-matched peers on NVML/DCGM telemetry; flags the degraded unit in **6 minutes** versus a multi-day ticket loop.
- Co-authoring two manuscripts targeting SIGMETRICS 2027: a five-fleet GPU failure taxonomy (~50,000 GPUs: Titan, Summit, Marconi100, and others; permutation null on every claim) and a detection-method paper built on the blind H100 validation.
- Packaged for fleet deployment: Helm chart, Go Kubernetes node-labeler controller (client-go) reconciling per-GPU health labels, multi-arch images on GHCR, and end-to-end CI on k3d; install verified from a clean cluster.

Undergraduate Researcher (Incoming), Justice-Centered Edge AI

JDEI-funded research, Cal Poly (advisor: ex-Intel silicon-validation faculty)

Fall 2026

San Luis Obispo, CA

- Selected for funded research building privacy-preserving image/video processing on the NVIDIA Jetson Orin Nano for food-equity applications; on-device inference so sensitive footage never leaves the hardware.

Electrical Lead, Avionics & Edge-AI Hardware

Poly UAS (Cal Poly Unmanned Aerial Systems, AUVSI SUAS team)

Apr 2026 – Present

San Luis Obispo, CA

- Own the aircraft's electrical system end to end: custom avionics board set (power/integration carrier, CAN backplane) and NVIDIA Jetson edge-AI carrier/sensor hardware in KiCad, under flight size, weight, and power constraints, enabling onboard real-time computer vision.

Lab Operations Intern, Systems & Automation

Natera

Jun 2025 – Sep 2025

Pleasanton, CA

- Reduced validation and packaging cycle time from ~20 min to 10–12 min (>40%) by mapping the workflow and removing redundant verification on a high-throughput production line.
- Automated barcode validation and pathology-report retrieval in Python; wrote root-cause documentation adopted team-wide.

TECHNICAL PROJECTS

- **RISC-V CPU (RV32I) on FPGA** (SystemVerilog, Vivado): multi-cycle full-ISA CPU from the RTL up (10-op ALU, FSM control, 32×32 register file); synthesized and verified on a Basys-3 through waveform-level debug.
- **Audio ADC → DAC signal chain on custom PCB** (AVR, LTspice): designed, simulated, and fabricated a binary-weighted 4-bit DAC on a 2-layer board with register-level firmware; 62.5 mV/step within ±1 LSB across all 16 codes.
- **theta-probe thermal sensor node** (ESP32-C3, KiCad): designing v1 of a PCB series toward a standalone thermal-resistance instrument for GPU research; three-sensor 1-Wire bus, humidity sensing, and I2C expansion headers.

TECHNICAL SKILLS

Languages: Python, C/C++, SystemVerilog/Verilog, TypeScript, Bash, MATLAB

GPU & Systems: NVML, DCGM, GPU thermal/power characterization (T4–H100), Prometheus, SLURM, Kubernetes (Helm, client-go), Linux, Git, permutation testing, survival analysis, scikit-learn, pandas, NumPy

Hardware & Embedded: NVIDIA Jetson, STM32, AVR, I2C/SPI/UART, register-level I/O, PCB design (KiCad, Autodesk Fusion), LTspice, RTL/FPGA (Vivado, static timing), oscilloscope/DMM, SMD/reflow soldering

LEADERSHIP & ADDITIONAL

President, Poly Engineering Consulting: lead client-facing engineering project teams

Sep 2024 – Present

Lab Assistant, Cal Poly EPIC: teach embedded systems to precollegiate students

Summer 2026

Shipped software: two production web platforms in live use (Stripe ticketing with QR check-in; consulting-ops platform) 2026